DIGITALMODEL · PIPELINE COMPANY BROCHURE PACK · 2026-05-07

CLIENT COLLATERAL · WALL THICKNESS GTM BROCHURE

Pipeline wall thickness screening that turns a 72-case multi-code study into a fast pilot conversation.

Headline: 72 cases, DNV-ST-F101/API RP 1111/PD 8010-2, 6 pipe sizes, X65, 500m water depth.

7.62 mm to 23.87 mm minimum wall thickness envelope identified across the published comparison set.

72

parametric cases in the underlying results set

6

pipe sizes from 6 in to 20 in

3

code bases compared side by side

500 m

water depth basis with X65 steel and 1.0 mm corrosion allowance

Client-use note. This brochure is derived from the existing demo dataset and report artifacts in the repository. It is suitable for prospect-facing GTM discussion and pilot scoping, but it is not a substitute for project-specific wall thickness design, certified material data, installation loading definition, or formal code compliance sign-off.

Executive hook

Pipeline teams often need a quick, defensible view of how code choice changes required wall thickness before they commit to a deeper design cycle. This digitalmodel brochure packages an existing multi-code comparison into a client-ready narrative that shows where thickness requirements move, what tends to govern, and how a pilot could be framed.

In the reference dataset, the published minimum-wall-thickness summary spans **7.62 mm to 23.87 mm** across **6 in, 8 in, 10 in, 12 in, 16 in, and 20 in** pipe sizes. The comparison covers **DNV-ST-F101**, **API RP 1111**, and **PD 8010-2** at **10, 15, 20, and 25 MPa** internal pressure for **X65** at **500 m water depth**.

The value proposition is simple: give a pipeline company a transparent screening pack they can review in one meeting, then convert interest into a tightly scoped pilot with auditable inputs and reproducible outputs.

DATASET SIGNAL

24 / 72

candidate result rows in the raw sweep are marked unsafe, which demonstrates why automated sizing and code-to-code comparison matter before a client locks an envelope.

TYPICAL GOVERNING PATTERN

Propagation-led

Propagation or propagation buckling governs most rows in the existing data, with pressure/burst/hoop checks governing only a small subset at higher pressure.

What the pack does

1 · COMPARES CODES

Shows how the three cited code bases change minimum required wall thickness for the same size, grade, depth, and pressure basis.

2 · SURFACES GOVERNING CHECKS

Highlights which checks govern the result set so engineering and business stakeholders can discuss where conservatism enters the envelope.

3 · FRAMES A PILOT

Converts a demo analysis into a prospect-facing pilot package with clear inputs, outputs, and limitations rather than vague digital-twin marketing claims.

Client discussion outcomes enabled by this brochure

- Rapid screening of whether code basis materially shifts wall thickness and tonnage envelope.
- Early alignment on which pipe sizes and pressure points deserve detailed engineering follow-up.
- Clear separation between reusable automation and the project-specific verification still required for final design.

Sample outputs

The table below reproduces representative minimum-wall-thickness outputs from the existing JSON summary. In this dataset, API RP 1111 is the lowest thickness result for each listed size, while PD 8010-2 is the highest by approximately 0.95 mm to 1.00 mm over the same sizes.

PIPE SIZE	OD (MM)	DNV-ST-F101 MIN WT (MM)	API RP 1111 MIN WT (MM)	PD 8010-2 MIN WT (MM)	OBSERVED SPREAD	TYPICAL GOVERNING CHECK
6 in	168.3	8.51	7.62	8.57	0.95 mm	Propagation / propagation buckling
8 in	219.1	10.79	9.90	10.90	1.00 mm	Propagation / propagation buckling
10 in	273.1	13.19	12.30	13.30	1.00 mm	Propagation / propagation buckling
12 in	323.9	15.47	14.58	15.58	1.00 mm	Propagation / propagation buckling
16 in	406.4	19.14	18.31	19.31	1.00 mm	Propagation / propagation buckling
20 in	508.0	23.65	22.87	23.87	1.00 mm	Propagation / propagation buckling

ILLUSTRATIVE INSIGHT

The raw result set includes **48 safe rows** and 24 rows flagged unsafe at candidate wall thicknesses in the sweep, with the largest published utilisation in the dataset occurring for a **20 in DNV-ST-F101** case at **15.1 mm** wall thickness.

SOURCE REFERENCES USED

/mnt/local-analysis/workspace-hub/digitalmodel/examples/demos/gtm/results/demo_02_wall_thickness_results.json

/mnt/local-analysis/workspace-hub/digitalmodel/examples/demos/gtm/output/demo_02_wall_thickness_report.html

/mnt/local-analysis/workspace-hub/digitalmodel/examples/demos/gtm/media/demo_02_wall_thickness.gif

/mnt/local-analysis/workspace-hub/digitalmodel/examples/demos/gtm/media/demo_02_wall_thickness_workflow.gif

Methodology / code basis

This brochure is grounded in the existing repository artifacts rather than new calculations. The data source identifies the demo as a **wall thickness multi-code comparison** with these explicit metadata values:

72 total cases | 6 pipe sizes | DNV-ST-F101 | API RP 1111 | PD 8010-2 | 10/15/20/25 MPa | 500 m water depth | X65 | SMYS 448 MPa | SMTS 531 MPa | 1.0 mm corrosion allowance

How to read the outputs

- The summary values shown in this brochure are the minimum published wall thicknesses per pipe size and code from the existing demo results.
- The raw results include governing-check labels such as propagation buckling, propagation, pressure containment, burst, and hoop stress.
- Because this is brochure collateral, the emphasis is on transparent comparison and pilot framing rather than reproducing every intermediate equation or load case in the full workflow.

Why this matters commercially

A pipeline company does not need a generic AI pitch. It needs a clear view of whether automated multi-code screening can shorten concept selection, narrow design envelopes faster, and hand off a traceable basis to detailed engineering. This pack is designed to support that conversation.

Pipeline-company pilot deliverables

- **Client-specific input register:** nominated diameters, design pressures, grade assumptions, corrosion allowance, depth basis, and any required load-case notes.
- **Automated comparison workbook or JSON export:** auditable case matrix with code-by-code minimum wall thickness and governing checks.
- **Management-ready PDF/HTML pack:** summary pages for engineering, project, and procurement stakeholders.
- **Engineering handoff note:** explicit list of items that remain project-specific, including material data confirmation, fabrication tolerances, installation loads, collapse inputs, and final compliance review.
- **Pilot closeout review:** a short session showing what the automation resolved quickly and where detailed project engineering still governs the final answer.

Limitations / caveat

- This brochure does **not** claim final wall thickness design, code compliance certification, or fitness-for-purpose approval for any real project.
- The content is limited to the repository dataset and referenced media/report artifacts cited above.
- Actual project execution would require verified inputs, project-specific load definitions, fabrication and installation criteria, material certificates, and formal engineering review under the client’s approved procedures.
- The raw sweep includes both safe and unsafe candidate rows; that pattern should be treated as evidence of screening logic, not as a direct recommendation to use any single unsafeguarded thickness value.

Prepared as self-contained digitalmodel client collateral for the pipeline-company brochure pack. No external fonts, scripts, or network dependencies are required to view this file.